

Jarret Dlugy

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EDUCATION

University of Massachusetts Amherst, GPA: 3.97/4.0

Expected Graduation: December 2026

Bachelor of Science in Mechanical Engineering

Relevant Coursework: Manufacturing Processes, Wind Turbine Design, Design of Mechanical Components, System Dynamics, Fundamentals of Electrical Engineering, Material Science, Intro to MATLAB, ME Lab, Statistics

WORK EXPERIENCE

Mechanical Engineering Intern, Lexington Medical Inc, Bedford, MA

May 2026 - August 2026

- Contributed to fast paced R&D engineering team with the company's first design verification for new product line of medical staplers by executing or supporting 9 of 15 testing protocols
- Designed, prototyped, and assembled a robust test fixture assembly on an aggressive timeline, to transform a previously labor-intensive, non-repeatable test method, improving test repeatability and measurement consistency
- Evaluated and compared critical device characteristics using qualitative and quantitative metrics such as CT scanners, comparators, vision systems, and physical gages
- Leveraged optical systems in a laboratory setting to generate quantitative data for a poorly characterized product defect lacking an objective inspection method
- Investigated stapler performance in correspondence with quantitative defect data to define acceptance criteria
- Prototyped and evaluated multiple in line defect detection approaches including optical sensing, pneumatic displacement sensors, as well as custom mechanical measurement methods to be used during product inspection
- Designed and developed an optical sensor based detection fixture for controlled environment quality control inspection of all subsequent products within new product development
- Coordinated with cross functional teams to ensure conformance with quality control and inspection standards

Battery Engineer Intern, Amel Energy, Amherst, MA

May 2025 - August 2025

- Assembled lithium-ion coin cells for performance evaluation of battery materials and life cycle testing
- Manufactured electrodes with varying chemistries to find sustainable and non-toxic carrier solvent and polymer binder for lithium-ion battery manufacturing
- Designed, 3D-modeled, and 3D-printed a custom 24-bed battery testing rack to optimize previously unorganized testing enclosure layout, improve accessibility to coin cell testing channels, and increase battery testing capacity
- Conducted first-cycle charge/discharge testing and analyzed performance data in written report delivered to COO
- Created comprehensive SOP documents for key lab processes including coin cell assembly and electrode manufacturing to assist in the training of new employees
- Designed and 3D printed custom spacers and brackets to optimize the layout of battery testing units
- Communicated results directly to COO and CEO both verbally and in a written final report

RELEVANT PROJECT EXPERIENCE

Drivetrain Team Co-Lead, UMass Amherst Mass Mileage Club, Amherst, MA

September 2025 - May 2026

- Manage sub team of 4 engineering students to minimize losses in Mass Mileage vehicle drivetrain system for international competition with the objective of achieving highest MPG
- Reverse engineered OEM drivetrain components to leverage proven design features accelerating development
- Designed and prototyped crankshaft adapter, using skills such as SolidWorks, 3D printing, and CAD Drawings, to supplement new easy to use and reliable centrifugal clutch while working around safety and design constraints
- Overseeing additional projects involving lightening, increasing workability, and reducing vibration of drivetrain

PCB Mounting Assembly Term Project

September 2025 - December 2025

- CNC machining team lead, creating aluminum mounts as part of larger PCB mounting assembly
- Parts were designed for manufacturing conforming to tool, time, and functional constraints given by professor
- Aluminum mounts prototyped using 3D printing and then manufactured on 2 axis CNC mill with custom programming and workpiece set up

SKILLS

Software: SolidWorks, Excel, MATLAB, Microsoft Word, PowerPoint, PDM/PLM, SolidWorks Drawing/Drafting

Manufacturing/Prototyping: CNC Mill, Lathe, 3D printing, GD&T, DFMA, General Machine Shop Tooling, Soldering

ACTIVITIES AND INTERESTS

Strength Training, All Season Hiking, Running, Travel (Hungary, Vietnam, Switzerland, Bosnia, Cambodia, Chile)